



*Energy Harbor Nuclear Corp  
Perry Nuclear Power Plant  
10 Center Road  
Perry, Ohio 44081*

**Rod L. Penfield**  
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September 28, 2023  
L-23-192

10 CFR 50.73(a)(2)(iv)(A)

ATTN: Document Control Desk  
U. S. Nuclear Regulatory Commission  
Washington, DC 20555-0001

SUBJECT:  
Perry Nuclear Power Plant  
Docket No. 50-440, License No. NPF-58  
Licensee Event Report Submittal

Enclosed is Licensee Event Report (LER) 2023-003, "Plant Trip During Reactor Protection System Power Transfer." There are no regulatory commitments contained in this submittal.

If there are any questions or if additional information is required, please contact Mr. Glendon Burnham, Manager – Regulatory Compliance, at (440) 280-7538.

Sincerely,

A handwritten signature in black ink, appearing to read "Rod L. Penfield", written over a horizontal line.

Rod L. Penfield

Enclosure:  
LER 2023-003

cc: NRC Project Manager  
NRC Resident Inspector  
NRC Region III Regional Administrator

Enclosure  
L-23-192

LER 2023-003

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollections.Resource@nrc.gov](mailto:Infocollections.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: [oir\\_submission@omb.eop.gov](mailto:oir_submission@omb.eop.gov). The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

**1. Facility Name**

Perry Nuclear Power Plant



050



052

**2. Docket Number**

00440

**3. Page**

1 OF 4

**4. Title:**

Plant Trip During Reactor Protection System Power Transfer

**5. Event Date**

Month: 08, Day: 10, Year: 2023

**6. LER Number**

Year: 2023, Sequential Number: 003, Rev No.: 00

**7. Report Date**

Month: 09, Day: 28, Year: 2023

**8. Other Facilities Involved**Facility Name: ☐ 050, Docket Number: ☐ 052**9. Operating Mode**

1

**10. Power Level**

100

**11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)****10 CFR Part 20**☐ 20.2203(a)(2)(vi)**10 CFR Part 50**☐ 50.73(a)(2)(ii)(A)☐ 50.73(a)(2)(viii)(A)☐ 73.1200(a)☐ 20.2201(b)☐ 20.2203(a)(3)(i)☐ 50.36(c)(1)(i)(A)☐ 50.73(a)(2)(ii)(B)☐ 50.73(a)(2)(viii)(B)☐ 73.1200(b)☐ 20.2201(d)☐ 20.2203(a)(3)(ii)☐ 50.36(c)(1)(ii)(A)☐ 50.73(a)(2)(iii)☐ 50.73(a)(2)(ix)(A)☐ 73.1200(c)☐ 20.2203(a)(1)☐ 20.2203(a)(4)☐ 50.36(c)(2)☒ 50.73(a)(2)(iv)(A)☐ 50.73(a)(2)(x)☐ 73.1200(d)☐ 20.2203(a)(2)(i)**10 CFR Part 21**☐ 50.46(a)(3)(ii)☐ 50.73(a)(2)(v)(A)**10 CFR Part 73**☐ 73.1200(e)☐ 20.2203(a)(2)(ii)☐ 21.2(c)☐ 50.69(g)☐ 50.73(a)(2)(v)(B)☐ 73.77(a)(1)☐ 73.1200(f)☐ 20.2203(a)(2)(iii)☐ 50.73(a)(2)(i)(A)☐ 50.73(a)(2)(v)(C)☐ 73.77(a)(2)(i)☐ 73.1200(g)☐ 20.2203(a)(2)(iv)☐ 50.73(a)(2)(i)(B)☐ 50.73(a)(2)(v)(D)☐ 73.77(a)(2)(ii)☐ 73.1200(h)☐ 20.2203(a)(2)(v)☐ 50.73(a)(2)(i)(C)☐ 50.73(a)(2)(vii)☐ OTHER (Specify here, in abstract, or in NRC Form 366A).**12. Licensee Contact for this LER****Licensee Contact**

George Dujanovic, Staff Nuclear Engineering Specialist, Regulatory Compliance

**Phone Number (Include area code)**

(440) 280-5200

**13. Complete One Line for Each Component Failure Described in this Report**

| Cause | System | Component | Manufacturer | Reportable to IRIS | Cause | System | Component | Manufacturer | Reportable to IRIS |
|-------|--------|-----------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
|       |        |           |              |                    |       |        |           |              |                    |

**14. Supplemental Report Expected**☒ No☐ Yes (If yes, complete 15. Expected Submission Date)**15. Expected Submission Date**

Month

Day

Year

**16. Abstract** (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On August 10, 2023, at 0039, with the reactor in Mode 1 and 100% rated thermal power, an automatic Scram occurred caused by the loss of both Reactor Protection System (RPS) busses while transferring the power supply on RPS B bus from Alternate B to Normal. This also caused an isolation signal to all the Main Steam Isolation Valves (MSIVs). High Pressure Core Spray (HPCS) and Reactor Core Isolation Cooling (RCIC) systems automatically initiated, due to lowering water level in the reactor vessel, to restore and control water level in the reactor vessel. Reactor vessel pressure control was accomplished by cycling the Safety Relief Valves (SRVs). The RPS Power Transfer Switch was replaced, and the reactor was restarted on August 15, 2023.

The direct cause of the reactor trip was inadvertent overtravel of the RPS power transfer switch, which interrupted power to both RPS busses. The root cause of the reactor trip is attributed to a failure to implement a design change for the RPS power transfer switch to prevent recurrence of a plant scram due to a known design vulnerability.

The RPS Power Transfer Switch was replaced with a like-for-like switch as an immediate corrective action, and a planned corrective action is to upgrade the switch to remove the known design vulnerability.

This event is reportable due to the actuation of RPS, including a reactor scram, isolation of the MSIVs, and the actuation of Emergency Core Cooling Systems (ECCS) under 10 CFR 50.73(a)(2)(iv)(A).





**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

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|                                                   |                                         |                               |               |                               |                    |
|---------------------------------------------------|-----------------------------------------|-------------------------------|---------------|-------------------------------|--------------------|
| 1. FACILITY NAME<br><br>Perry Nuclear Power Plant | <input checked="" type="checkbox"/> 050 | 2. DOCKET NUMBER<br><br>00440 | 3. LER NUMBER |                               |                    |
|                                                   | <input type="checkbox"/> 052            |                               | YEAR<br>2023  | SEQUENTIAL<br>NUMBER<br>- 003 | REV<br>NO.<br>- 00 |

**NARRATIVE**

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

**EVENT DESCRIPTION:**

On August 10, 2023, at 0039, with the plant operating in Mode 1 and approximately 100 percent power, a reactor scram occurred during the transfer of the Reactor Protection System (RPS) [JC] B power supply [JX] from the alternate power supply to the normal power supply. During the transfer, a loss of both RPS A and B busses [BU] occurred, causing the Main Steam Isolation Valves (MSIVs) [ISV] to isolate, and a reactor scram. Safety Relief Valves (SRVs) [RV] automatically opened to lower and maintain pressure in the reactor vessel [RCT], due to the loss of the Condenser [COND] resulting from the MSIV closures. Water level in the reactor vessel dropped to Level 2 (130 inches above the reactor core), and High Pressure Core Spray (HPCS) [BG] and Reactor Core Isolation Cooling (RCIC) [BN] automatically initiated to restore and maintain water level in the reactor vessel. The plant responded to the unexpected RPS actuation and reactor scram as designed.

The Control Room Operator was transferring the RPS B power supply from the B alternate feed transformer [XFMR] to the normal B motor-generator [MG] set, in accordance with the operating procedure, following restoration from RPS breaker maintenance. The transfer of RPS power is a non-paralleling "break before make" transfer, causing the RPS bus loads to be momentarily de-energized. A Division 2 half-scam is expected during the transfer. However, an unexpected break in power to the RPS A bus also occurred, providing the second half-scam signal and a full RPS actuation. The RPS power switch operation was suspect and was subsequently replaced. The reactor was restarted on August 15, 2023.

To perform a transfer from the B alternate transformer to the normal B MG set, the RPS power transfer switch [JS] is rotated from the 45-degree position (ALT B) to the 0-degree position (NORM). Similarly, to transfer the RPS bus A power supply from the normal A MG set to the A alternate transformer, the power transfer switch is rotated from the 0-degree position (NORM) to the 315-degree position (ALT A).

The operating procedure contains a caution statement that reads, "Movement of the MG SET TRANSFER switch 1C71-S1 shall be performed in a slow and steady motion. When placing it in NORM, 3 to 5 degrees of over-travel may cause loss of RPS bus A." This caution statement was added due to a similar incident that occurred in 1988.

**INVESTIGATION:**

As part of the root cause investigation, vendor testing was performed on the removed switch and did not identify any functionality issues. It was determined that the RPS A or B bus power break occurs when the power transfer switch over travels by approximately 10 degrees from the NORM position, in either direction.

A review of internal operating experience identified that the RPS power transfer switch has a latent design error that allows for overtravel of the switch and subsequent de-energization of both RPS busses. The switch design places operators in an error likely situation by not eliminating the human factor from the switch manipulation. Overtravel of the RPS power transfer switch is possible and has resulted in a previous plant scram (1988).



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## NARRATIVE

A review of external operating experience shows that plants with the same switch design have experienced scrams due to slight overtravel when placing the RPS power transfer switch from the ALT to NORM position. There were four events from 1985 to 1989. Identified causes were personnel error and design limitation of the switch. Corrective actions included switch inspection, operator training, placing a caution tag on the switch, and adding a caution to the operating instructions. There were also two events from 1993 to 1994. Identified cause was overtravel of the switch. Corrective actions consisted of replacing the switch, installing a warning placard on the switch, and considering a design change of the switch.

## CAUSE OF EVENT:

## Direct Cause:

The direct cause of the scram was attributed to inadvertent overtravel of the Reactor Protection System (RPS) power transfer switch, resulting in a second switch contact associated with RPS A opening, causing the scram.

## Root Cause:

The root cause of the scram is failing to implement higher-level safety precedence sequence corrective actions, such as a design change implementation, to prevent recurrence of a plant scram due to a known design vulnerability.

## CORRECTIVE ACTIONS:

## Completed Actions:

The RPS power transfer switch was conservatively replaced by a like-for-like switch by a work order as a corrective maintenance activity, prior to start up from the forced outage.

## Scheduled Actions:

The RPS power transfer switch design will be upgraded during the next scheduled refueling outage, in the Spring of 2025, to prevent a similar event from occurring again.

## EVENT ANALYSIS:

A Probabilistic Risk Assessment (PRA) bounding evaluation was performed for the August 10, 2023, automatic scram with loss of condenser. An analysis of this plant scram results in a delta Core Damage Frequency (CDF) and a delta Large Early Release Frequency (LERF) that are well below the acceptable thresholds of 1.0E-06/yr and 1.0E-07/yr, respectively, as discussed in Regulatory Guide 1.1.74. The risk of this event is therefore considered to be of very low safety significance in accordance with the Regulatory Guidance.

In reference to NEI 99-07, Indicator Guideline, this scram has been determined be an unplanned scram with complications.

This event is reportable due to the actuation of RPS, including a reactor scram, isolation of the MSIVs, and the actuation of Emergency Core Cooling Systems (ECCS) under 10 CFR 50.73(a)(2)(iv)(A).



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**NARRATIVE**

**PREVIOUS SIMILAR EVENT:**

A similar event occurred on June 18, 1988, as documented in LERs 88-025-00 and 88-025-01. Overtravel of the RPS power transfer switch occurred while transferring the switch from Alternate A to Normal, resulting in deenergizing both A and B RPS busses. This event was attributed to personnel error, with a contributing factor of a poor human factors design. A caution sign was installed near the switch, which was later removed, warning of the possibility of switch overtravel causing a de-energization of the opposite RPS bus. A similar caution was incorporated into the operating procedure. LER 88-025-00 mentions a design change is being considered for the RPS power transfer switch but was then later removed in supplemental LER 88-025-01.